

HOME AUTOMATION Using Arduino Through Android Device

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This project is based on Android app and Arduino Uno using Bluetooth as the wireless communication medium. It is a simple and flexible home automation system with only a few electronic components, without complex soldering. This project is designed to control three appliances, but it can be extended to six or more.

Circuit and working

The block diagram of the home automation system for controlling three

electrical appliances is shown in Fig. 1 and its complete circuit diagram in Fig. 2. It is built around Arduino Uno R3 (Board1), HC-05 Bluetooth module, three relays (RL1 through RL3) and a few other components. This project demonstrates how to turn on/off the appliances connected with the relays using an Android app on a smartphone or tablet.

Relay pin connections

Relay connections marked as x, y, COM (common), NO (normally open) and NC (normally connect) are shown in Fig. 3. x and y are two points across the coil of the relay.

The relay board consists of relay drivers.

PARTS LIST

Semiconductors:

- Board1 - Arduino Uno R3
- D1-D3 - 1N4007 rectifier diode
- LED1-LED3 - 5mm LED
- T1-T3 - BC547 npn transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R6 - 1-kilo-ohm

Miscellaneous:

- CON1 - 6-pin connector for HC-05 Bluetooth module
- CON2-CON4 - 3-pin connector for load
- BATT.1 - 12V battery
- RL1-RL3 - 12V, single-changeover relay

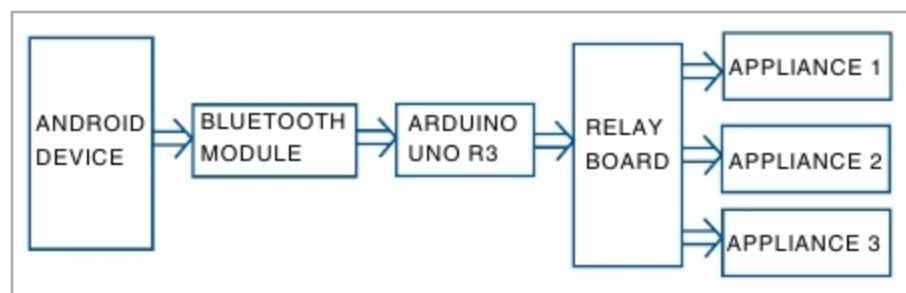


Fig. 1: Block diagram of home automation system



Fig. 3: Relay connections

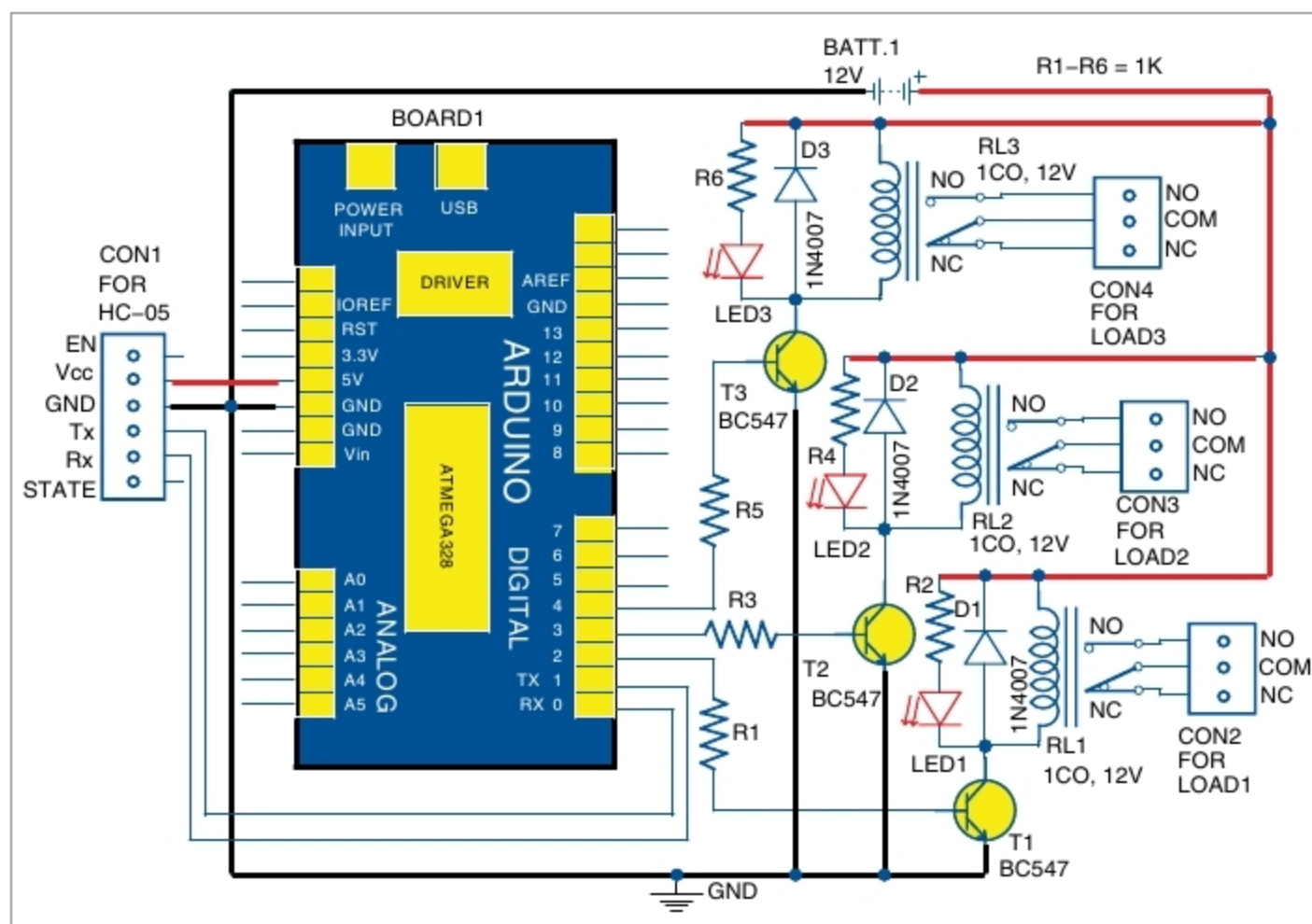


Fig. 2: Circuit diagram of home automation system

Three BC547 transistors (T1, T2 and T3) are used as relay drivers for controlling the three appliances.

Pin 2 of Arduino Uno is connected to T1 through base current-limiting resistor R1. Similarly, pins 3 and 4 of Arduino Uno are connected to T2 and T3 through R3 and R5, respectively.

All emitters of the transistors should be connected to ground (GND) pin of Arduino Uno.

Connect V_{CC} and GND pins of HC-05 Bluetooth module to 5V pin and GND pin of Arduino Uno, respectively. And, connect Tx and Rx pins of Bluetooth module to Rx and Tx pins of Arduino Uno, respectively.

Connect NO terminal of each relay to one terminal of